

Incubation Device to Study Brain Organoids with High-Field Magnetic Resonance Imaging

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Abstract

Brain Organoids are three-dimensional models derived from human pluripotent stem cells that more accurately replicate early human brain development than traditional models. As a result, they are invaluable tools for studying neurodevelopmental and neurodegenerative diseases. To study neural activity and the health of tissues in these models, researchers commonly use imaging techniques such as functional MRI (fMRI) and sodium MRI. However, a major limitation of high-field MRI imaging is the inability to maintain organoid viability during extended imaging sessions. Organoids require extremely specific conditions, including temperature regulation and nutrient supply, which are not supported within current MRI environments. This mismatch restricts the duration and quality of imaging, preventing accurate study of live organoids' function over time. To address this gap, we designed and tested an MRI-compatible incubation chamber capable of sustaining organoid viability under imaging conditions. The system integrates continuous media perfusion via a peristaltic pump, a temperature-controlled water bath to maintain ~37 °C, MRI-safe materials to prevent imaging interference, and an optical fiber port for optogenetic stimulation. Organoid viability was assessed using metabolic assays and calcium imaging to evaluate cellular activity over time. Results demonstrated that organoids maintained higher metabolic activity and functional calcium signaling in the chamber compared to non-regulated conditions, indicating improved physiological stability. These findings suggest the proposed system successfully supports organoid viability during simulated MRI conditions and can potentially enable longer, high-quality imaging of live brain organoids. This platform can enhance the study of human brain function and disease progression in a controlled imaging environment.

Introduction and background

Brain organoids are three-dimensional (3D) structures derived from human pluripotent stem cells (hPSCs) that replicate key aspects of early human brain development. Compared to traditional two-dimensional cell cultures and animal models, organoids provide a more physiologically relevant platform for studying human-specific biology. They mimic features such as layered cellular organization, formation of neurons, and early network formation, enabling researchers to investigate processes that are otherwise difficult to observe in vivo. Because organoids can also be derived from patient-specific cells, they offer a powerful tool for modeling neurodevelopmental and neurodegenerative diseases, including conditions such as microcephaly and Alzheimer's disease. As a result, brain organoids have become increasingly important in both basic neuroscience research and translational medicine, particularly for studying disease mechanisms and assessing potential therapies^{1,2}.

To study neural activity and tissue health in these models, researchers commonly rely on advanced imaging techniques such as functional magnetic resonance imaging (fMRI) and sodium (²³Na) MRI. fMRI measures brain activity indirectly by tracking changes in blood oxygen levels, providing insight into how different parts of the brain are functioning. In contrast, sodium MRI directly measures sodium ion concentration within tissue, which is closely tied to cellular metabolism, homeostasis of ions, and membrane health. Sodium gradients are essential for neuronal excitability, as they drive action potentials through ATP-dependent Na⁺/K⁺ pumps. Disruptions in these gradients can indicate early-stage dysfunction, metabolic stress, or loss of cellular viability. When used together, the information obtained from these imaging methods is complementary, enabling a more complete understanding of both functional and physiological processes within brain organoids³.

Despite these advances, a major limitation of high-field MRI imaging is the inability to maintain organoid viability during extended imaging sessions. Brain organoids require tightly controlled environmental conditions to remain viable, including stable temperature (approximately 37°C), continuous nutrient delivery through media perfusion, and appropriate gas exchange to maintain cellular metabolism⁴. Without these conditions, organoids rapidly lose physiological function, exhibit abnormal neural activity, and may undergo structural degradation. This leads to unreliable or distorted experimental data, limiting the interpretability of imaging results. The challenge becomes even harder in high-field MRI systems, where the small size of organoids results in low signal-to-noise ratios (SNR), requiring longer scan durations to obtain meaningful data. As a result, maintaining viability over extended imaging periods represents a major obstacle in obtaining accurate, high-resolution measurements from live organoids⁴.

While prior work has explored incubation systems for organoid culture and live imaging in optical or benchtop environments, these approaches are not compatible with high-field MRI systems. MRI environments impose strict physical and electromagnetic constraints, including limited bore diameter, sensitivity to motion, and the requirement for non-magnetic and non-conductive materials. Additionally, any integrated system must avoid interference with radiofrequency (RF) signals and gradient fields to preserve image quality. Current incubation platforms, which often rely on components containing metal, bulky heating systems, or open fluid reservoirs, cannot be directly translated into this setting⁵. Even when partial solutions exist, they typically fail to provide the full combination of temperature control, media perfusion, and MR compatibility required for prolonged imaging. Consequently, no existing system effectively maintains organoid viability while simultaneously supporting high-field MRI imaging, revealing a clear engineering and scientific gap in the field⁵.

This project addresses this gap by developing an MRI-compatible incubation chamber capable of maintaining organoid viability during extended imaging sessions. The system integrates several key features within a compact and MR-safe configuration. Continuous media perfusion is achieved using a peristaltic pump and

extended tubing to enable nutrient delivery while maintaining a safe distance from the imaging region. Temperature regulation is accomplished through a water bath system that maintains physiological conditions and distributes heat evenly to the organoid environment. In addition, the chamber is constructed using MRI-safe materials to prevent magnetic interference and ensure compatibility with high-field imaging systems. The overall design is engineered to fit within the MRI bore while maintaining stability and minimizing motion artifacts.

By preserving physiological conditions during imaging, the finalized system enables longer-duration scans and improves data quality. This represents a major improvement over current approaches, as it allows for the observation of live organoids under conditions that more closely reflect their natural state. In doing so, the device has the potential to expand the capabilities of neuroimaging by enabling measurements that were previously not feasible, particularly in small, low SNR biological samples. Ultimately, this work helps contribute to the study of human brain development and disease, providing a platform that can advance future research in neuroscience, drug development, and personalized medicine.

Methods

1. Model Design and Production

1.1 Chamber Design and Assembly

A custom MRI-compatible chamber was designed on SolidWorks to fit within a 6 cm MRI bore diameter. The chamber was prototyped using PLA from fused-deposition modelling (FDM) 3D printers found in the EPIC lab, chosen for its MRI compatibility and rapid and accessible modelling. The finalized prototype was made from resin using stereolithography (SLA) 3D printing for tighter tolerances and to prevent leakage from the water bath due to its water resistance. Three parts were assembled: (1) a cylindrical structure designed to pass the media tubing and hold the optical fiber clamps, (2) an optical fiber clamp at the top of the cylinder to hold the optical fiber, and (3) a base structure with a holder for a plexiglass tube that holds the organoid. The base was also designed to have an internal network of tubing for the circulation of heated water for temperature regulation. Silicone pads were adhered onto the optical fiber clamp surfaces to provide a suitable hold onto the fibers, and the clamp assembly was constructed using a $\frac{3}{4}$ " #2-56 machine screw as a pivot and a $\frac{1}{2}$ " #2-56 machine screw to secure the clamp. For complete MRI compatibility, these metal screws will be replaced with equivalent-sized MRI-compatible nylon screws. All the components were assembled for structural stability and a leak-free experiment.

1.2 Media Circulation System

Media circulation was completed using 2.5 m of tubing to maintain a minimum distance of 1.25 m from the MRI bore. A peristaltic pump was used to maintain a continuous flow of media, preventing contamination and allowing for nutrient exchange. Inlet and outlet ports were integrated into the chamber design and sealed to prevent leakage during media flow.

1.3 Temperature Regulation

A temperature-controlled water bath system was developed to maintain the required 37 °C within the chamber. Heating was completed using a network of tubing embedded at the base of the chamber, allowing warm water to circulate beneath the plexiglass tube containing the organoid. The tubing embedded in the base of the chamber was designed to concentrate heat delivery to the plexiglass. This encouraged an upward heat distribution through the water bath. A thermistor temperature circuit was added and coded via Arduino to monitor water temperature in real time. The thermistor used was a GA10K3A1A NTC Thermistor 10K bead⁶. A calibrated thermometer was used as a control to validate the temperature measurements from the thermistors.

1.4 Temperature Testing and Calibration

The temperature of the water bath was measured by placing the thermistors within the bath along with the control thermometer. Prior to data collection, each thermistor was calibrated by recording its output voltage at known reference temperatures and converting these voltages to temperature using a calibration curve. This allowed raw voltage readings to be accurately translated into temperature values. Measurements were recorded over time to evaluate the system's ability to maintain a constant temperature. The thermistor readings were compared to the control thermometer to verify their consistency. Adjustments to the algorithm were made to prevent fluctuations and maintain the target temperature of 37 °C. Data was analyzed using MATLAB.

Once the thermistor was calibrated, we assessed whether the temperature from the water bath was consistent throughout the entire system, including inside the MR tube and within the media. Thermistors were placed in both the water bath where the media tubes lay and inside the MR tube. Temperature measurements were recorded over time to evaluate the system's ability to maintain a stable temperature of ~37 °C. This step was necessary to verify that the organoid experienced uniform temperature conditions during incubation. Data was analyzed using MATLAB.

1.5 MRI Compatibility Testing

MRI compatibility will be assessed by placing the chamber in the scanner and completing imaging at Martinos Center for Biomedical Imaging, accompanied by Dr Xin Yu. Signal strength and the presence of artifacts will be assessed to confirm that the chamber materials do not interfere with the MRI measurements.

2. Organoid viability

2.1 Organoid and Chamber Preparation

Organoids and media were provided by Dr. Ella Zeldich from BU Medical School and used as supplied. A 4% (w/v) agarose solution (prepared in HBSS) is cast in a petri dish and allowed to solidify. On the day of the experiment, a section of agarose was cut and shaped to fit within the central region of the MR tube. A well was created in the agarose using a modified pipette tip to securely place the organoid. The Organoid was transferred into the agarose well using a sterile transfer pipette, ensuring stable positioning⁷. The agarose-encapsulated organoid in the MR tube is positioned within the 3D-printed prototype chamber for imaging and incubation.

2.2 Simulating a Physiologically Stable Environment

The organoid embedded in the MR tube will be maintained in the chamber for 5 hours to simulate extended MRI imaging. The chamber is placed into its temperature-controlled water bath maintained at 37 °C, with temperature continuously monitored using a thermistor. Using the chamber's media inlet and outlet ports, controlled perfusion at under 2.5 $\mu\text{L}/\text{min}$ ⁸ will maintain continuous flow throughout all 5 hours of incubation. The chamber contains approximately 250 μL of media surrounding the organoid, meaning the entire volume is exchanged roughly every 100 minutes at this flow rate. This higher exchange rate improves nutrient delivery and waste removal while maintaining low shear stress, supporting organoid viability during extended imaging. Variables included temperature (37 °C), media composition, agarose concentration, and chamber configuration, which will remain constant and controlled throughout the experiment. Measurements are obtained at baseline, 2.5 hours, and 5 hours. A positive control Organoid is maintained under standard culture conditions (37 °C with media) without extended chamber exposure.

2.3 Viability Assessment

To assess the organoid's viability, we performed a PrestoBlue⁹ metabolic assay and calcium imaging to assess how metabolically active the organoid is and whether it maintains cellular activity. Four conditions were tested: (1) test sample (organoid in chamber at 37 °C), (2) positive control (organoid at 37 °C in incubator), (3) negative control (organoid at room temperature), and (4) media-only baseline. Each sample contained 500 μ L total volume with a 9:1 media: PrestoBlue ratio. Metabolically active cells reduce resazurin to resorufin, resulting in an increase in absorbance (~570-600 nm) as an indicator of viability¹⁰. Absorbance was measured using a NanoDrop every 20 minutes from 0 to 100 minutes. For each measurement, 2 μ L of media was placed on the pedestal, and the instrument was blanked between each time point to ensure accurate readings¹¹. An increase in absorbance over time indicates metabolic activity, which we can conclude that the Organoid has maintained viability. The test sample was evaluated by comparing its trend to the positive control, while confirming that the negative control and media baseline remained low, validating that signal changes were due to viable organoid activity. Values were normalized to the positive control to determine relative organoid viability. This procedure was repeated with a new set of organoids and media to ensure consistency and accuracy.

For calcium imaging, following incubation under simulated MRI conditions, organoids expressing GCaMP8s were imaged directly without the need for exogenous dyes. Imaging was performed using a two-photon microscope, maintaining physiological conditions at 37 °C. Time-lapse fluorescence images were collected to capture transient changes in intracellular calcium. Fluorescence intensity was analyzed over time by selecting regions of interest within the organoid, and calcium activity was quantified as $\Delta F/F_0$, representing intracellular calcium fluctuations and overall cellular activity.

2.4 Data and Statistics

Data were analyzed in MATLAB by plotting absorbance vs. wavelength from NanoDrop measurements for each sample over time. Multiple time points (0–100 min) were used to determine the slope of 570nm absorbance change, which reflects metabolic activity. The slope for each sample was compared to assess relative organoid viability. Additionally, calcium imaging data ($\Delta F/F_0$) were analyzed to evaluate cellular activity, where transient fluorescence changes indicate active signaling, further supporting organoid viability.

Results

Optical Fiber Clamp Testing

Iterations of the optical fiber clamp assemblies were tested to validate ease of use, secure holding, and durability. The first prototype of the clamp was designed with too much clearance between the silicone pads, so it was unable to hold the optical fiber securely when pulled. The second iteration of the clamp was able to hold the fiber securely under significant tension, but lacked alignment features to keep it centered in the assembly, posing a risk of damaging the fiber. The final iteration securely held the optical fiber under tension, which will allow for proper positioning during testing, included alignment features to readily center the fiber, and was durable enough to withstand repetitive tightening and loosening without screw thread stripping.

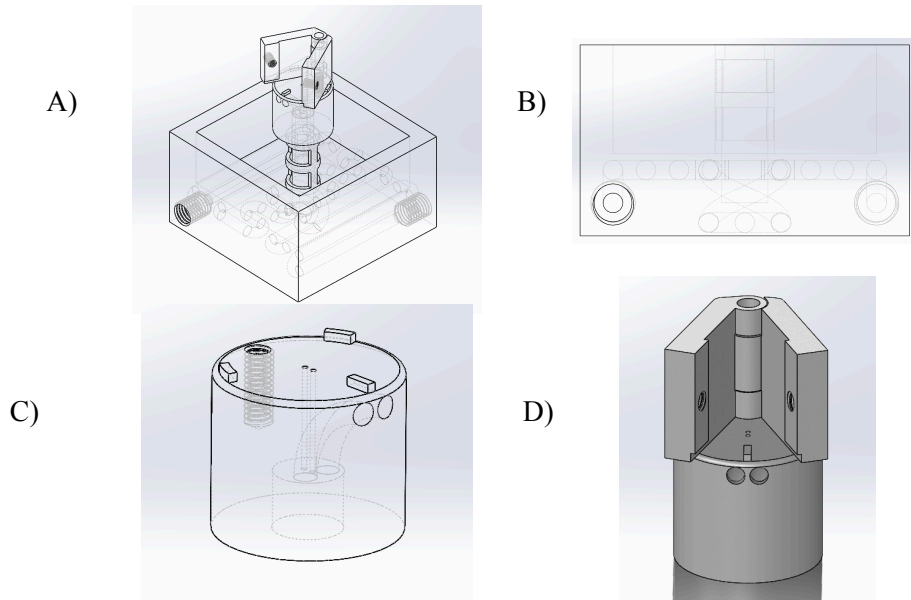


Figure 1: (A) Full chamber assembly showing integrated water bath and internal components. (B) Side view of the water bath with internal tubing layout visible. (C) Tube cap with media and optical fiber interface. (D) Optical fiber clamp assembly for securing and positioning the fiber.

Media Circulation Testing

Testing of the peristalsis pump to be used for media perfusion demonstrated adequate volumetric flow. Different tube diameters and wall diameters were tested, and the smallest diameter tube able to be used was 3/16" OD 1/32" ID. Using this diameter tubing and moderate peristalsis pump flow rate, a volume of ~400μL was successfully maintained inside the MRI tube.

Heating Bath Testing

Several iterations of heating baths were tested. Successful designs included a base with channels throughout, intended to maintain a certain temperature in the organoid tube and to heat media before perfusion (Figure 1A). This design proved to be promising, but the initial testing failed due to the unreliability of fused-deposition modeling (FDM) 3D printing. The FDM print did not properly seal each individual PLA layer and showed signs of leakage between layers. Future iterations done with SLA printing circumvented this issue, and the water bath was able to steadily maintain temperatures inside the MRI tube of $\sim 37.3 \pm 0.5^\circ\text{C}$. MRI tube temperatures can be controlled using an external thermistor, which maintains an average temperature of $\sim 38.1 \pm 0.7^\circ\text{C}$ after heating (Figure 2).

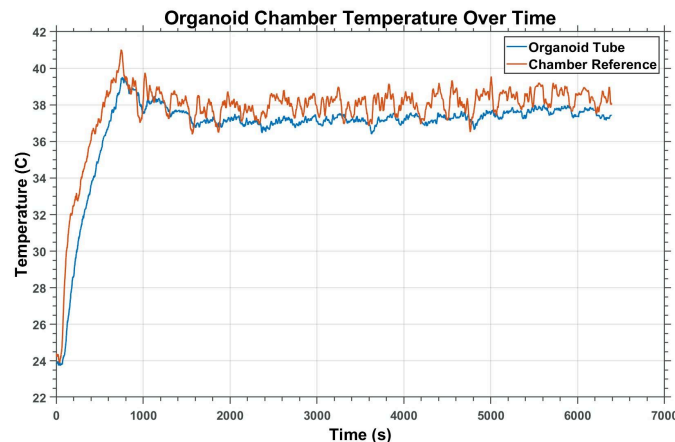


Figure 2: Measured temperature inside the organoid tube compared with reference temperature of the thermistor placed in the chamber.

Metabolism Testing Results

All tested organoids showed increased 570nm absorbance over time, indicating that metabolism took place (Figure 3). The absorbance of the water bath yielded a slope of $4.64 \times 10^{-4} \text{ min}^{-1}$, the bench organoid yielded a slope of $0.54 \times 10^{-4} \text{ min}^{-1}$, the test organoid yielded a slope of $6.40 \times 10^{-4} \text{ min}^{-1}$, and the slope of the media control was $0.29 \times 10^{-4} \text{ min}^{-1}$. This confirms that both the positive control and test chamber resulted in significant organoid metabolism, with the organoid in the MRI tube exceeding that of the organoid kept in a 37°C water bath. Additionally, the organoid kept at room temperature only showed moderate metabolic activity, with a slope larger than that of the media control, but still an order of magnitude lower than the positive control or the MRI chamber, proving the MRI chamber represents an important part of continued organoid functioning (Figure 4).

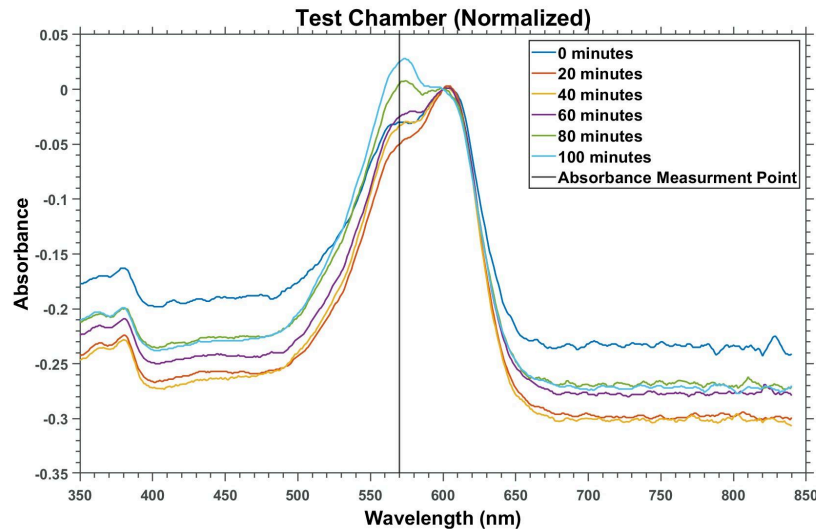


Figure 3: Absorbance data for organoid tested in MRI chamber. Absorbances were normalized to 600nm and measured at 570nm.

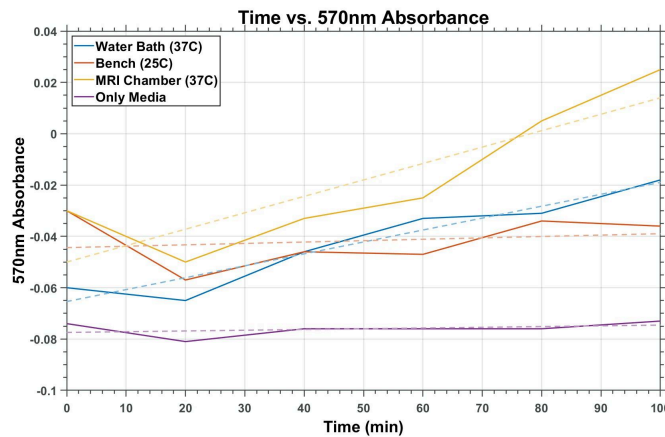


Figure 4: 570nm Absorbance over time for several test conditions: organoid kept at 37°C in a water bath, organoid kept at 25°C, organoid kept in the MRI chamber at 37°C, and only media as a control. Dashed lines represent the linear fit of the corresponding data.

Calcium Imaging

Fluorescent imaging of the brain organoid demonstrated excitability of neurons tagged with channelrhodopsin. Figure 5 shows the results of calcium imaging for a single organoid, with excitable neurons labeled. Figure 6 shows the results of many optogenetic stimulation trials, where it can clearly be seen that the organoid consistently exhibits detectable neuronal activity after optogenetic stimulus (Figure 5F).

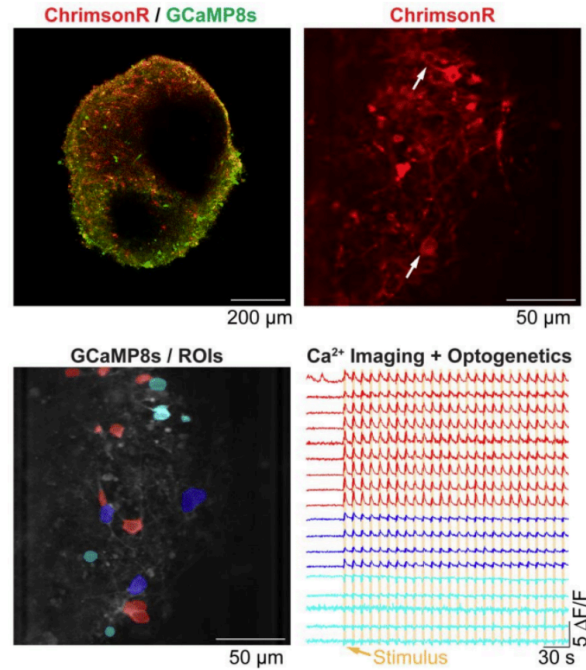


Figure 5: Dorsal cortical organoids were transduced with AAVs for neuron-specific expression of ChrimsonR and GCaMP8s. White arrows point to cells with clear membrane localization of ChrimsonR (top right). During two - photon imaging of GCaMP8s, a fiber-coupled 561-nm laser was used to activate ChrimsonR, leading to stimulus-evoked neuronal excitation and intracellular Ca^{2+} influx visualized through an increase in GCaMP8s fluorescence (red, strongly responding cells, blue, weakly responding cells, cyan, non-responding cells)

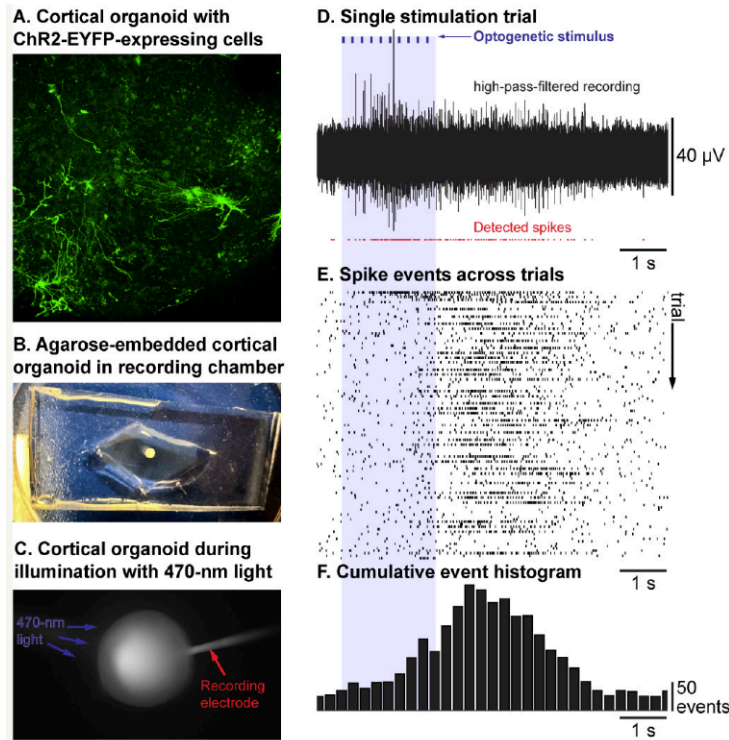


Figure 6: Setup and data from optogenetic stimulation of a brain organoid under a two-photon microscope. (A) Fluorescent image of a section of an organoid with cells expressing ChR2-EYFP; (B) Image of the chamber the organoid is tested in; (C) Image of organoid with recording electrode; (D) Detected neuronal activation spikes in a single trial with stimulus event labeled; (E) Detected neuronal activations across many trials; (F) Histogram of cumulative events taking place at a certain time during the trials

Discussion

The primary design goal of this project was to develop an MR-compatible incubation platform capable of sustaining human brain organoids during prolonged, high-field neuroimaging. As established in current live-imaging literature, longitudinal MRI studies of brain organoids have historically been restricted by the inability to maintain environmental controls inside the MRI, leading to rapid degradation. The experimental results validate that the custom-engineered chamber successfully overcomes this barrier by meeting all mechanical and thermal requirements. Through physical testing, the integrated SLA printed water bath system successfully warmed the internal chamber, while the leaks that were previously observed in early prototypes were avoided. Using external thermistors, the internal MRI tube environment was actively monitored, demonstrating the system's ability to regulate heat at $37.3 \pm 0.5^{\circ}\text{C}$. Furthermore, iterative testing of the optical fiber clamp yielded a design that securely holds the fiber under continuous tension. This structural stability is a defining metric of success, as maintaining perfectly stable physical conditions is required not just to prevent motion artifacts during MRI scanning, but to allow for simultaneous optogenetic targeting.

With the thermal and mechanical properties stabilized, the system's biological compatibility was evaluated using a PrestoBlue metabolic assay. The 570nm absorbance data confirmed that the organoids incubated within the custom chamber over extended periods exhibited metabolic activity. Notably, the metabolic rate of the organoid in the chamber trended higher than that of the positive control maintained in a standard 37°C water bath, while the room temperature and media only negative controls remained predictably low. This is significant, as it demonstrates that the custom chamber's thermal environment successfully provides a well regulated environment, ensuring that brain organoids maintain metabolic viability over extended periods outside of a traditional culture setting.

Calcium imaging of the brain organoids demonstrated the feasibility of calcium optogenetics as a positive control for future experiments. In future trials, optogenetic fibers will be held close to the organoid using the designed optical fiber clamp and these fibers will rapidly stimulate GCaMP8 and detect the overall fluorescent response of the organoid, allowing for detection of neuronal activation. By stimulating and measuring the response of GCaMP8-tagged neurons during MRI testing, these future experiments will be able to quantify neuronal activity and compare this Ca^{2+} data with data from fMRI.

Despite the overall success of this project, the scope of the validation was restricted by resource constraints. First, access to brain organoid cells was restricted as producing them is costly, limiting the statistical power and the number of viability tests that could be performed. Second, testing the chamber inside the actual MRI scanner itself was precluded, as the MRI facility is undergoing construction.

This project successfully engineered an incubation chamber that maintains the conditions required for live brain organoid imaging. By ensuring structural stability for both MRI and optical microscopy, the chamber lays a strong foundation for future studies. Future iterations could also incorporate pH and CO_2 monitoring to address other environmental factors. Additionally, once the facility is operational, the chamber will be tested inside the MRI to further validate its use.

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